Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Scalable Distributed Systems · Real-Time Fraud Detection · ML-Powered Anomaly Detection · Event-Driven Architecture · AWS · Python · Observability

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PROFESSIONAL SUMMARY

Software Engineer with 8 years owning scalable distributed backend systems in production - including fraud-adjacent income verification pipelines that processed 1.6M+ housing compliance records and enforced data integrity across regulated financial workflows. At Entrata, built real-time event-driven architectures (RabbitMQ async processing, retry logic, idempotency guarantees) and observability tooling that reduced triage time 40% at 99.5% uptime across 100K+ monthly active users. Built AgentProbe - an ML-powered evaluation platform with anomaly detection, latency monitoring, and automated quality gating - and conducted anomaly detection research (Isolation Forest) on large-scale datasets at George Mason University. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Engineering & Backend: Python, SQL, PHP, JavaScript, TypeScript, Bash, FastAPI, Docker, Microservices, Scalable Distributed Systems, Event-Driven Architecture, AWS (S3, EC2, RDS, Glue), Redis, RabbitMQ, REST APIs, CI/CD (GitHub Actions), Real-Time Processing, Low-Latency Systems, PostgreSQL, MySQL, Oracle, Linux, Git

Fraud & ML: Fraud Detection, Anomaly Detection (Isolation Forest), ML-Powered Detection Pipelines, Machine Learning, PyTorch, scikit-learn, NLP, BERT, PySpark, Feature Engineering, Statistical Modeling, Graph-Based Analysis, Data Integrity Validation, Observability, Monitoring

AI & Agents: LangGraph, LangChain, Agentic AI, RAG Pipelines, LLM Orchestration, ChromaDB, Model Evaluation, Hallucination Detection, HITL, Prompt Engineering, OpenAI API, Anthropic API, FastAPI

EXPERIENCE

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML-powered anomaly detection pipelines (Python, R) applying Isolation Forest and statistical modeling across large-scale WHO and UK Road Safety datasets - $R^2 = 0.977$; full pipeline from data ingestion through feature engineering, model optimization, and anomaly validation across 190+ countries.
- Mentoring 600+ students in Python and database design; designed monitoring dashboards improving research comprehension by 40%.

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built scalable agentic backend with real-time validation guardrails - conditional routing blocking non-compliant outputs before delivery; ML-powered detection of constraint violations across distributed data sources.
- Implemented RAG pipeline with low-latency optimization and monitoring on retrieval quality and response groundedness.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned fraud-adjacent income verification pipelines end-to-end** - processed 1.6M+ HUD/Tax Credit financial records with multi-stage data integrity validation, anomaly detection, and reconciliation controls; zero compliance violations across 7 years; 100% PHI compliance on AWS infrastructure (Libsodium-encrypted pipelines, zero security incidents).
- **Designed scalable distributed backend architecture** - real-time event-driven processing via RabbitMQ (async message routing, retry logic, idempotency guarantees, dead-letter handling); Docker microservices redesign (50% scalability improvement, 30% deployment reduction); Redis caching for high-performance low-latency data access at 100K+ monthly active users.
- **Built observability and monitoring tooling** - internal debugging infrastructure reducing triage time 40%; real-time performance monitoring across multi-tenant production environments; 99.5% uptime SLA across 120+ Agile sprints.
- **Cloud infrastructure at production scale** - AWS S3, EC2, RDS; 200+ SQL query optimizations (30% performance gain); scalable data processing pipelines transforming fragmented federal data into 180,000 domain records per release cycle.
- Mentoring 5+ junior engineers; 300+ peer code reviews; engineering excellence across 20+ full-stack production modules.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management backend modules (PHP, Python, PostgreSQL); fraud-adjacent data pipeline for 1.6M HUD/Tax Credit records with validation and integrity checks; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: ML-Powered Detection & Observability Platform Fraud Detection · ML

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Scalable ML-powered detection and monitoring platform: 15-endpoint FastAPI service, 5-metric anomaly detection engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated quality gate blocking regressions before merge, LLM-as-judge scoring, and multi-provider routing for low-latency optimization. Real-time monitoring and alerting across 1,240+ test cases with full observability into model behavior.

LeaseI: Agentic AI Backend with Fraud-Detection Controls Agentic · ML

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Distributed AI backend with 8-agent stateful graph, conditional routing, and multi-source data ingestion - HITL escalation when model confidence is below threshold, analogous to fraud decisioning escalation. 93.6% precision, 87.9% F1 across 1,240 data-intensive test cases; real-time monitoring with telemetry-driven quality scoring.

AI on FHIR: NLP-Driven Medical Records Query System NLP · Backend

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Scalable NLP backend with graph-based relationship traversal across FHIR resources - reduced retrieval time ~60%; secure RBAC architecture with full data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017